MIL-C-24643/20B
3 October 1986
SUPERSEDING
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8 November 1985
MIL-C-24643/20
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LS7SGU (INCLUDING VARIATION LS7SGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper, conductor, uncoated, class B stranding (see table I for size).
Second - Extruded silicone rubber insulation (see table I for diameter).
Third - Glass braid.
Fourth - Braid covering. Standard identification code applied by method 1.
Fifth - The seven conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly. Cabling sequence shall be consecutive starting with no. 1 from center outward.
Sixth - Binder or combination binder and barrier.

Construction variant for type LS7SGU

Seventh - Cross-linked polyolefin jacket. (See table I for thickness.) Cable surface marking required.

Construction variant for type LS7SGA

(B) Seventh - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking not required.
Eighth - Braided metal armor and paint.

(B) denotes changes.

AMSC N/A FSC 6145
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TABLE I. Details.

Military 1/ part no. M24643/20	Size 2/	Conductor size AWG	Type's overall diameter (inches)				Diameter over insulation nominal (inch)	Diameter over braid covering maximum (inch)	Jacket thickness nominal (inch)	Conductor resistance per 1000 feet at 25°C maximum (ohms)
			LS7SGU		LS7SGA					
			min	max	min	max				
-01UN	3	16(class B)	---	0.545	---	0.595	0.096	0.130	0.040	4.3
-02UN	4	14(class B)	---	.595	---	.645	.112	.143	.040	2.55

1/ Unarmored. When armored option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS7SGU and LS7SGA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	(see table I)
Voltage withstand - volts, root mean square, minimum....	3000
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage.....	No failure
Abrasion resistance - scrapes, minimum.....	250
(B) Crack resistance.....	No damage
Group B:	
Gas flame - 1 hour.....	No failure
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Cold bending, cable - (maximum mandrel diameter 12 times maximum cable diameter).....	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lbf/in^2, minimum.....	750
Elongation - percent, minimum.....	125
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure
Armor (type LS7SGA only) - conformance to material construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:
All sizes 2000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-20)

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